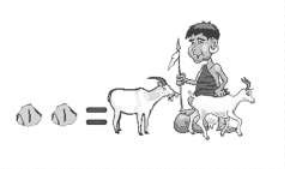
Tema 1. Introducción

Fundamentos de Computadores – G.I.T.I.

Luis Felipe Romero Gómez, 2020

Contenido

1. Historia de la computación
2. Arquitectura de Von Neumann
3. Ciclo de Instrucción
4. Generaciones de computadores



1.1 Breve referencia histórica

- …cuando los dedos no eran suficientes
  - Ábaco, Babilonia, 3000-2000 A.C
  - China, 1000- A.C.
- Calculadoras "a pedales" (ruedas dentadas)
  - Anthykithera (100 A.C.)
  - Napier, 1612: Los *huesos* de Napier
  - Schickard, 1623: Sumas y restas. Desbordamiento
  - Pascal,1642: *Pascaline*. Sumas y restas.
  - Leibnitz, 1671-1694: Multiplicación, división, raíces
  - Jaquard, 1801: Telares programables con tarjetas

Breve referencia histórica

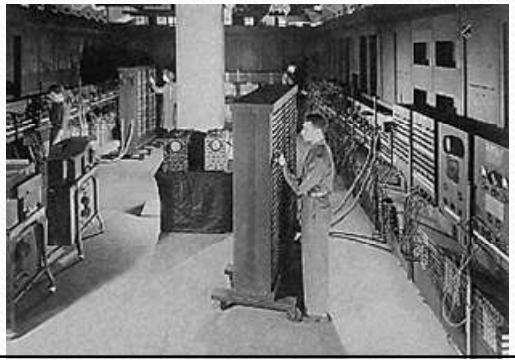
- Charles Babbage, 1812-1842:
  - Máquina de diferencias
    - *No construida, sólo descrita "en papel"*, 1812
  - Ada Lovelace Byron, primera programadora
  - Máquina analítica,1842
    - Tamaño previsto 4.5m, a manivela o vapor
- Holleritz, 1887 y la IBM, 1924
- Leonardo Torres-Quevedo, 1914
  - Primera calculadora electromecánica, *"en papel"*
  - Relés, precursor de los transistores

Breve referencia histórica

- Boole (1854) y Shannon (1938)
  - Puertas lógicas elementales
- La máquina de Turing, 1937
  - Máquina programable de propósito general
  - *Colossus* y ENIGMA, 1943
- Atanassoff, 1939-1942
  - Inventor de los computadores electrónicos digitales
  - Primer diseño *"en papel"* (ABC)
  - Tecnología de válvulas
- Zuse, Alemania, 1939-1941
  - Z1 1935, 1938 (primer electromecánico, de relés)
  - Z3, 1939-1941 (primer electrónico digital, de válvulas)
- Aiken, 1939-1944
  - Mark 1. Computador electromecánico decimal

ENIAC (Mauckly y Eckert), 1942-1945

- Primer (?) computador electrónico digital
- Electrical Numerical Integrator and Calculator
- Cálculo de trayectorias balísticas
- 30 m de largo x 2.5 m de alto
- 150 KW, 18800 válvulas
- 100 Herzios
- (5000/357/38) (+*%)/segundo
- Cifras de 10 dígitos decimales
- Sin condicionales
- EDVAC, su gran sucesor

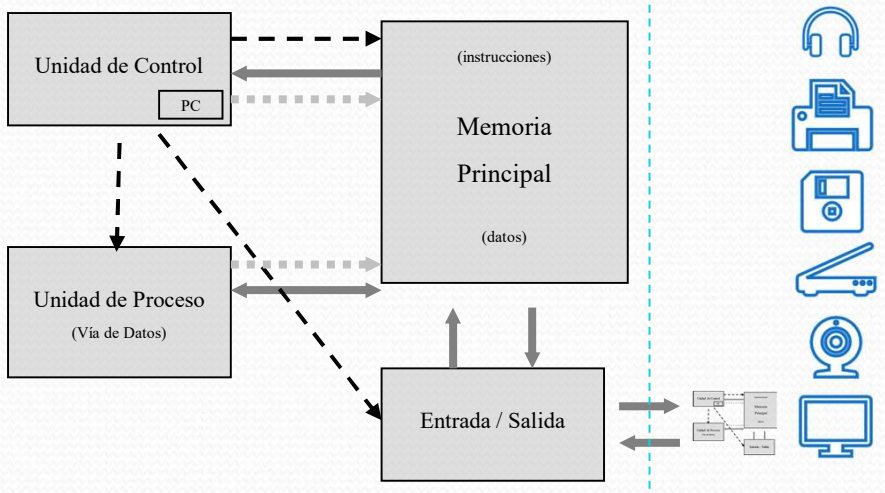


1.2 La Arquitectura de Von Neumann

Partes de un computador (con arquitectura Von Neumann):

- Unidad Central de Proceso
  - Unidad de control (genera señales de gobierno)
  - Unidad de proceso o "vía de datos" (procesa la información)
- Memoria (almacén de datos e instrucciones)
- Buses
- Entrada y Salida

La Arquitectura de Von Neumann



1.3 Ciclo de Instrucción

- Instrucción
  - Una orden elemental que el computador es capaz de ejecutar sin requerir un paso previo
  - Un programa es un conjunto de instrucciones
  - Las instrucciones se guardan en la memoria principal
- Contador de programa (PC), o *puntero* de instrucciones
  - Un registro que *indica* la posición en memoria de la instrucción que se ejecutará inmediatamente

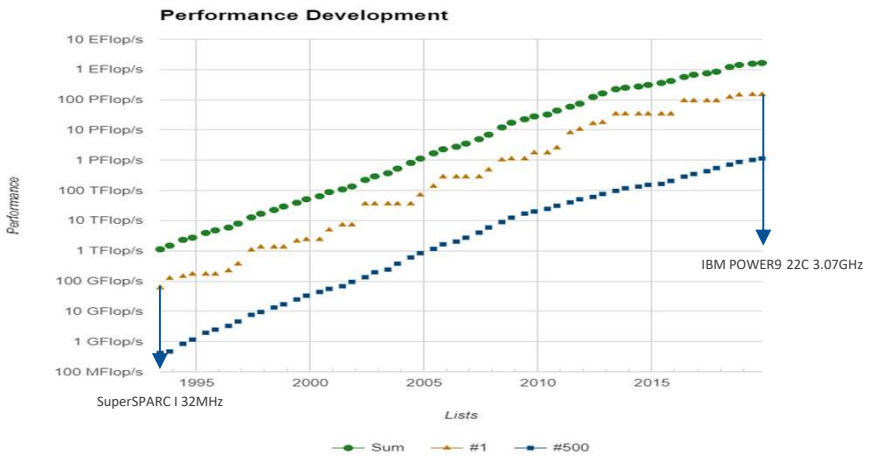
Ciclo de Instrucción

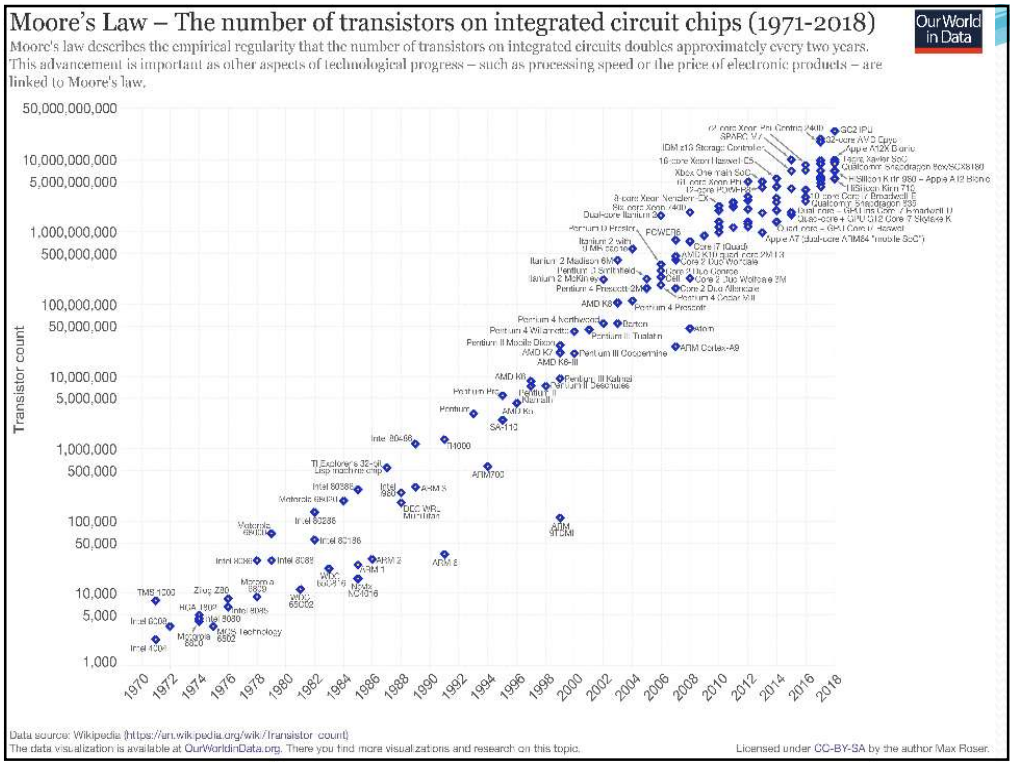
- Ciclo de instrucción
  - Búsqueda de instrucción (PC → MEM → UC)
  - Descodificación (UC)
  - Envío de señales para ejecución (UC)
    - Búsqueda de datos (MEM → VD)
    - Ejecución (VD)
    - Almacenamiento de resultados (VD → MEM)
  - Determinar próxima instrucción (cambiar PC)

Generaciones de computadores

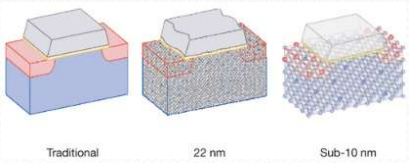
- Tubos de vacío 1940-1950
- Transistores 1950-1964
- Circuitos integrados 1964-1971
- Microprocesadores 1971-1985
- Arquitecturas RISC 1985-2002
- Arquitecturas multi-Core 2002-
- Próximas generaciones: ¿AI, QC?

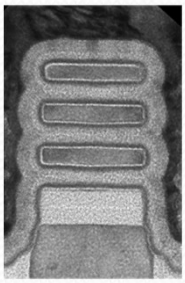
Rendimiento relativo

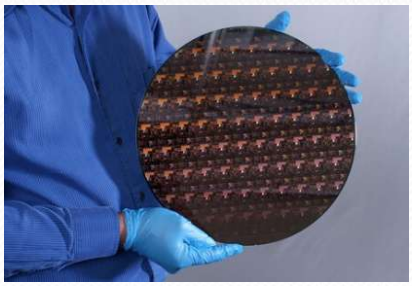


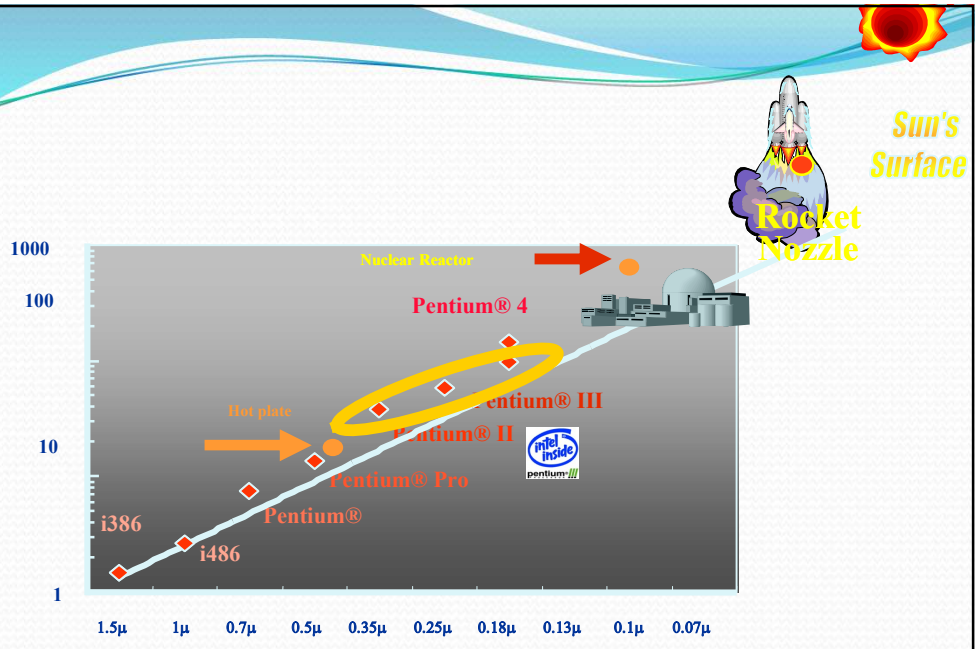


Integración







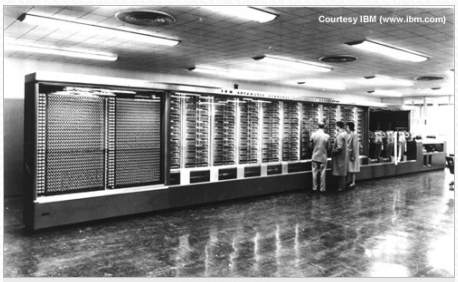


* "New Microarchitecture Challenges in the Coming Generations of CMOS Process Technologies" – Fred Pollack, Intel Corp. Micro32 conference key note - 1999.

TOP500 (noviembre 2021)

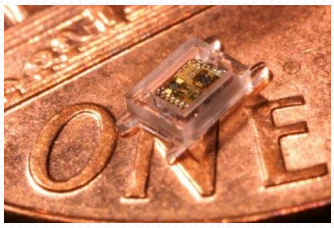
Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442,010.0	537,212.0	29,899
2	**Summit** - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	148,600.0	200,794.9	10,096
3	**Sierra** - IBM Power System AC922, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM / NVIDIA / Mellanox DOE/NNSA/LLNL United States	1,572,480	94,640.0	125,712.0	7,438
4	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway, NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
5	**Perlmutter** - HPE Cray EX235n, AMD EPYC 7763 64C 2.45GHz, NVIDIA A100 SXM4 40 GB, Slingshot-10, HPE DOE/SC/LBNL/NERSC United States	761,856	70,870.0	93,750.0	2,589

Las comparaciones son…



The Harvard Mark I

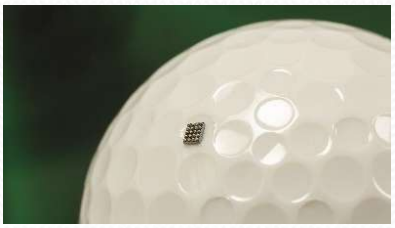
Micro Mote 1 (2011)



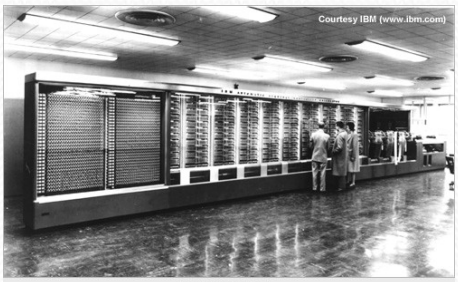
Las comparaciones son…

The Harvard Mark I

Kinetis KL02 (2013)

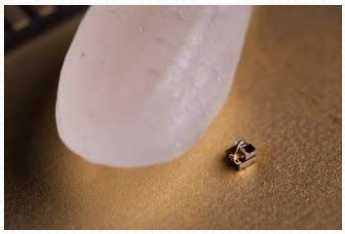


Las comparaciones son…



The Harvard Mark I

Micro Mote M3 (2018)

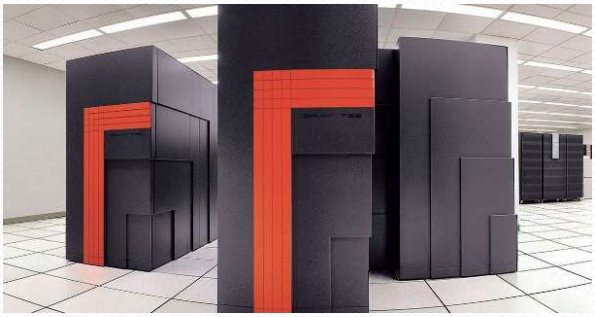


Las comparaciones son…

400 MFLOP (CRAY X-MP 1982)

400 MFLOP (Nexus 6, 2015)

Las comparaciones son…



CRAY T3E-900, 32GB, Edinburgh, 128GFLOP en 1997)

Xiaomi Mi 10, Snapdragon 865, 2,84 GHz, ~12GB, 128 GFLOP en 2020)

El futuro de los computadores

"This "telephone" has too many shortcomings to be seriously considered as a means of communication. The device is inherently of no value to us"
Western Union internal memo -- 1876.

"Who the hell wants to hear actors talk?"
H.M. Warner, Warner Brothers -- 1927

"The world potential market for copying machines is 5000 at most"
IBM to the founders of Xerox – 1959

"The average American family hasn't time for television"
The New York Times – 1939

"There will never be a mass market for motor cars - about 1,000 in Europe - because that is the limit on the number of chauffeurs available!"
Spokesman for Daimler Benz

"Man will never reach the moon regardless of all future scientific advances"
Dr. Lee De Forest, inventor of the vacuum tube -- 1957

"I think I may say without contradiction that when the Paris Exhibition closes, electric light will close with it, and no more will be heard of it"
Erasmus Wilson, Professor at Oxford University -- 1878

El futuro de los computadores

“Computers in the future may only have 1000 vacuum tubes and weigh no more than one-and-a-half tonnes."
- Popular Mechanics, 1949

"I think there is a world market for maybe five computers."
- Thomas Watson, Chairman of IBM, 1943

"I can assure you that data processing is a fad that won't last the year."
- Chief Business Editor, Prentice Hall, 1957

"Yeah, microchips, but what... is it good for?"
- an IBM senior enginner, 1968

"There is no reason anyone in the right state of mind will want a computer in their home."
- Ken Olson, President of Digital Equipment Corp, 1977.

"640k is enough for anyone, and by the way, what's a network?"
- William Gates III, President of Microsoft Corporation, 1984